

Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Principal Cloud Architect — Architecture Governance

Version: 1.2

Date: August 2026

Status: Accepted

Part of: ACRME Architecture Decision Records — one of four standalone, self-contained records.

About ADRs. An Architecture Decision Record captures a single significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original. This record is self-contained: it can be read without any companion document. Evidence tags: [Documented], [Decided], [Derived], [Assumed] (see Appendix).

ADR-002 — Quota and Capacity Management

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, Platform Engineering, FinOps

Related constraints: HC-2, HC-3, HC-7 (hard constraints)

Context

Each customer needs three CRGs per region (Prod, NonProd, DR). Azure tracks compute quota per subscription per region per SKU family; quota exhaustion silently blocks CR creation and VM deployment. The engine must:

- Prevent a NonProd surge from consuming quota that Prod CR creation needs (isolation).
- Make emergency capacity transfer (see ADR-003) **quota-neutral** where possible — i.e. reshuffle capacity without waiting hours for an Azure quota-increase approval during a crisis.
- Provide clean per-environment cost attribution for chargeback.
- Enforce a protected DR reservation that NonProd growth can never erode.

Azure **Quota Groups** (Preview) allow multiple subscriptions/CRGs to share a group-level quota pool, but Azure does **not** natively support intra-group sub-reservations — so any sub-partition of a group pool must be enforced by the engine.

Decision

Adopt a **Two-Quota-Group-per-region model with an engine-enforced DR floor**:

1. **Two groups per region:** one **Prod-only** group and one **shared NonProd+DR** group. [Decided]

$$\text{Prod_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_v})$$
$$\text{NonProd_DR_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_v})$$

2. **Engine-enforced DR floor** inside the NonProd+DR group (Azure has no native sub-reservation): [Decided]

$$\text{DR_Floor_vCPU}(R) = \text{potential_dr_demand}(R) \times \text{vCPU_per_instance} \times \text{dr_ratio_max}$$
$$\text{Effective_NonProd_Ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$$

The floor is always sized at **dr_ratio_max (0.40)** — the upper bound of the DR ratio range — so it never needs to grow if the operating ratio is later raised (0.30 → 0.40). Only actual Prod demand growth expands the floor. [Decided]

3. Hard-constraint enforcement at placement:

- **HC-2 CAPACITY_FLOOR** — CR headroom $\geq 2 \times \text{requested_vm_count} \times \text{SKU_vCPU}$.
- **HC-3 QUOTA_FLOOR** — env-type-aware group headroom check (Prod group / effective NonProd ceiling / DR headroom). Reads from the QuotaGroup entity, not legacy per-subscription QuotaRecord. [Decided]
- **HC-7 DR_FLOOR_INTEGRITY** — rejects NonProd placement that would push non-prod_quota_used above the effective ceiling, evaluated **after** HC-3. [Decided]

4. **Legacy per-subscription quota tracking is retained for monitoring/audit only** and superseded for all placement and scoring checks by group-level fields. [Decided]

5. **Continuous protection:** the reconciliation loop watches for floor violations and emits DRFloorViolationDetected; NonProd CRG scale operations that approach the ceiling are operator-gated.

Worked topology example: Prod group 128 vCPU; NonProd+DR group 80 vCPU; DR floor 32 vCPU → effective NonProd ceiling 48 vCPU.

Group Accounting Formulas (normative) All four are **engine accounting controls** — they do not create a native Azure sub-reservation: [Documented]

DR_Floor_vCPU	=	Potential_DR_Demand × vCPU_Per_Instance × DR_Ratio_Max
Effective_NonProd_Ceiling	=	NonProd_DR_Group_Limit - DR_Floor_vCPU
NonProd_Headroom	=	Effective_NonProd_Ceiling - NonProd_Used_vCPU
Group_Headroom	=	Group_Limit - Group_Used

Exact Enforcement Controls

- Every **NonProd** increase performs a group check **and** a subscription check. [Documented]
- Every **DR** increase performs a group check, subscription check, SKU check, capacity check, **and** active-incident check.
- A **separate detector** recalculates the DR floor from authoritative assignment/allocation data. Any disagreement between the command-time and detector calculations **disables automatic NonProd expansion** (fail-safe) and raises DRFloorViolationDetected.
- Group propagation is **polled** — no fixed propagation SLA is assumed.

groupType Preview Dependency The Quota-Group groupType property (AllocationGroup = advisory vs EnforcedGroup = enforced) is **preview-only** in the Azure Quota REST API and is not GA. [Documented] The engine's accounting controls are deliberately **engine-level and do not depend on groupType enforcement**. If EnforcedGroup is ever relied upon to drop the engine's own subscription-level check, that dependency must pass a preview-acceptance proof-of-concept and a recorded decision first. Engineering constraint: pin the exact preview API version in all quota-group calls and add version-drift detection to the platform health check.

Consequences

Positive: - Prod isolation is enforced at the Azure control-plane level — a NonProd surge cannot starve Prod. - Tier 2 Emergency Capacity Transfer becomes **truly quota-neutral** (ARM operations only, no Azure approval gate) because NonProd and DR draw from the same

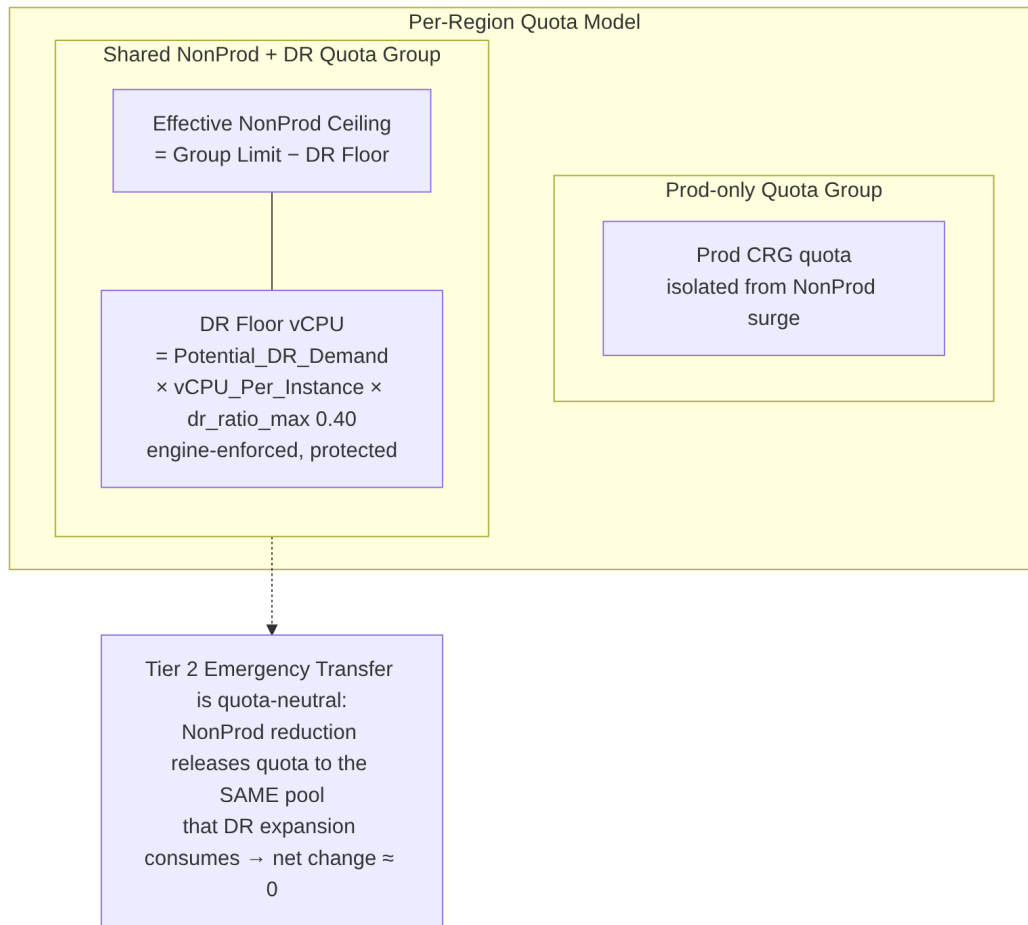


Figure 1: **Figure 1.** Two-quota-group-per-region model — an isolated Prod-only group and a shared NonProd+DR group with the engine-enforced DR floor and the effective NonProd ceiling.

group pool (see ADR-003). - Clean chargeback: one Azure billing record per group per region.
 - The DR floor guarantees a protected reservation NonProd can never erode.

Negative / trade-offs: - The DR floor is a **soft ceiling the engine must maintain continuously** — Azure won't enforce it. A bug could allow NonProd to breach the floor; mitigated by DRFloorViolationDetected alerting and operator gates. [Decided] - Sizing the floor at dr_ratio_max over-reserves quota slightly at lower operating ratios — a deliberate reliability-over-cost trade, reviewed quarterly. [Decided] - Quota_Score and HC-3 must be environment-type-aware (three code paths). - **Hard dependency on Azure Quota Groups GA** — validated by proof-of-concept before rollout; if groupQuotas returns 404 in the target tenant/region, escalate to Azure Support.

Alternatives Considered

Alternative	Why Rejected
Single shared quota pool (all three CRGs)	Breaks Prod isolation — a NonProd surge can consume Prod's quota. [Decided]
Three separate groups (one per CRG)	Maximum isolation but makes Tier 3 transfer non-atomic — releasing NonProd quota lands in the wrong group, forcing an Azure quota request mid-crisis (hours of RTO impact). [Decided]
Per-subscription quota only (legacy)	Cannot make emergency transfer quota-neutral without risky cross-subscription coordination. [Decided]
DR floor sized at current ratio	Undersizes the floor if the ratio is later increased, blocking DR expansion. [Decided]

Appendix — ADR Summary

ADR	Hard Constraints Applied	Key Open Items
ADR-002 Quota & Capacity	HC-2, HC-3, HC-7	Azure Quota Groups GA validated by proof-of-concept before rollout

Appendix — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR

Appendix — Evidence Tag Taxonomy

Tag	Meaning
[Documented]	Traceable to Azure platform behaviour or documentation
[Decided]	An explicit ACRME design choice recorded in this ADR set
[Derived]	A logical consequence of a documented constraint or decision
[Assumed]	Architectural judgement pending proof-of-concept validation

Related ADRs

- **ADR-001 — Region Selection** (acrme_adr_001_region_selection.md)
- **ADR-003 — Capacity Management during Disaster Recovery (DR)** (acrme_adr_003_capacity_management.md)
- **ADR-004 — Forecast and Increase of Capacity and Quota** (acrme_adr_004_forecast_and_increase.md)

Document Status: Accepted

Next Review: After proof-of-concept validation of Azure Quota Groups GA and quota-release latency, and on resolution of the consumer-credential model and engine-mode state-machine items.